

Log-concavity failures at a fixed fraction of the independence number

Research draft prepared for Jan Mikulík

27 September 2026

Abstract

We give a self-contained argument for an explicit sequence of trees of maximum degree 65 whose independence sequences fail log-concavity at indices k_h with $k_h/\alpha(T_h) \rightarrow \beta < 0.99$. The construction is a complete 64-ary tree with one pendant leaf added to each terminal vertex. At a fixed positive activity, a single pendant edge realizes exactly an unstable fixed point of the hard-core recursion. Conditional means remain well separated relative to conditional standard deviations, producing two separated mass concentrations and a local obstruction to log-concavity. The argument supplies a proposed affirmative solution to Galvin's Question 3.1. All proof steps are included; independent expert review and a broader priority check remain outstanding.

Statement and context. Let $i_k(T)$ be the number of independent sets of size k and let $\alpha(T)$ be the largest possible size. Galvin asks whether some fixed $c > 0$ permits arbitrarily large trees with

$$i_k(T)^2 < i_{k-1}(T)i_{k+1}(T), \quad k \leq (1-c)\alpha(T)$$

[1, Question 3.1]. The earlier boundary constructions do not provide such a fixed c . The preprint of Fang, Lu, Nevo, Yao and Zheng proves central log-concavity for sufficiently large forests, and leaves open whether its asymptotically universal upper endpoint b_{lc} equals 1 [2, Section 9]. The construction below gives $c = 1/100$ and $b_{lc} < 1$. It concerns log-concavity, not a counterexample to unimodality.

Fix $b = 64$. Let T_0 be an edge rooted at an endpoint. Obtain T_{h+1} by joining a new root to the roots of b disjoint copies of T_h . Equivalently, T_h is the complete b -ary tree of height h , with one extra leaf at each vertex on level h . Then

$$|V(T_h)| = b^h + \frac{b^{h+1} - 1}{b - 1}, \quad \Delta(T_h) \leq 65.$$

Theorem 1. *For every sufficiently large h , there is an interior index k_h such that*

$$i_{k_h}(T_h)^2 < i_{k_h-1}(T_h)i_{k_h+1}(T_h), \quad \frac{k_h}{\alpha(T_h)} \rightarrow \beta,$$

where, with $r \in (0, 1)$ the nonzero solution of $(1-r)(1+r)^{64} = 1$,

$$q = \frac{r}{1+r}, \quad \pi = \frac{q}{1+q}, \quad \beta = \frac{r + (1-r)\pi + \pi/63}{1 + 64/4095} < \frac{12350}{12477} < \frac{99}{100}. \quad (1)$$

Consequently Galvin's requested constant can be taken to be $c = 1/100$.

1. Exact realization of the fixed point. Put $\delta = 1 - r$. The equation for r becomes

$$\delta(2 - \delta)^b = 1.$$

There is a solution with

$$2^{-b} < \delta < 2^{1-b} \leq b^{-2}. \quad (2)$$

Indeed the left side is less than 1 at 2^{-b} and greater than 1 at 2^{1-b} , since $2(1 - 2^{-b})^b \geq 2(1 - b2^{-b}) > 1$. It is strictly increasing on this interval: its derivative has the sign of $2 - (b+1)\delta$. The function $(1 - r)(1 + r)^b$ first increases and then decreases on $[0, 1]$, so this is the unique nonzero solution.

Set $\lambda = r/(1 - r) > 0$ and sample an independent set I with probability proportional to $\lambda^{|I|}$. This uses the same activity at every vertex, with no imposed boundary configuration. For a rooted tree let R be the ratio of partition functions with its root occupied and unoccupied. For T_0 ,

$$R_0 = \frac{\lambda}{1 + \lambda} = r.$$

The usual conditioning identity gives

$$R_{h+1} = \frac{\lambda}{(1 + R_h)^b} = r.$$

The last equality follows from $\lambda = r(1 + r)^b$. Thus the root occupation probability of every T_h is exactly q . In particular,

$$\frac{49}{100} \leq \frac{1 - b^{-2}}{2} \leq q < \frac{1}{2}, \quad \rho := bq, \quad \rho - 2 \geq \frac{7b}{15} > 1. \quad (3)$$

2. Conditional moments. Write $X_h = |I|$. Conditional on root state $s \in \{0, 1\}$, let

$$\mu_h^s = \mathbb{E}(X_h \mid s), \quad V_h^s = \text{Var}(X_h \mid s), \quad D_h = \mu_h^1 - \mu_h^0, \quad W_h = \max(V_h^0, V_h^1).$$

Given an unoccupied root, the b child subtrees are independent copies of the unconditioned law. Given an occupied root, they are independent copies conditioned on an unoccupied child root. Therefore

$$\mu_{h+1}^0 = b((1 - q)\mu_h^0 + q\mu_h^1), \quad \mu_{h+1}^1 = 1 + b\mu_h^0, \quad (4)$$

$$V_{h+1}^0 = b((1 - q)V_h^0 + qV_h^1 + q(1 - q)D_h^2), \quad V_{h+1}^1 = bV_h^0. \quad (5)$$

The initial values are

$$\mu_0^0 = r, \quad \mu_0^1 = 1, \quad V_0^0 = r\delta, \quad V_0^1 = 0.$$

In particular

$$D_0 = \delta, \quad D_{h+1} = 1 - \rho D_h, \quad W_{h+1} \leq bW_h + \frac{b}{4}D_h^2. \quad (6)$$

Lemma 2. For every $h \geq 1$,

$$|D_h| \geq (1 - 1/(2b))(\rho - 2)^{h-1}, \quad W_h \leq \frac{5}{4b}D_h^2 = \frac{5}{256}D_h^2.$$

Moreover $|D_h|/b^h \rightarrow 0$.

Proof. By (2), $D_1 = 1 - \rho\delta \geq 1 - 1/(2b) > 1/2$. Whenever $|D_h| \geq 1/2$, $|D_{h+1}| \geq \rho|D_h| - 1 \geq (\rho - 2)|D_h|$. This proves the lower bound by induction. Also

$$W_1 \leq b\delta + \frac{b}{4}\delta^2 \leq \frac{1}{b} + \frac{1}{4b^3}, \quad \frac{W_1}{D_1^2} \leq \frac{5}{4b}.$$

For the induction step, (6) and (3) give

$$\frac{W_{h+1}}{D_{h+1}^2} \leq \frac{225}{49b} \left(\frac{W_h}{D_h^2} + \frac{1}{4} \right) \leq \frac{225}{49b} \left(\frac{5}{4b} + \frac{1}{4} \right) \leq \frac{5}{4b}.$$

The last scalar inequality holds at $b = 64$ (equivalently, $15525 < 15680$). Finally the exact solution

$$D_h = \frac{1}{1+\rho} + (-\rho)^h \left(\delta - \frac{1}{1+\rho} \right)$$

implies $D_h = O(\rho^h)$, and $\rho/b = q < 1$. □

3. A local obstruction to log-concavity.

Lemma 3. *Suppose a positive sequence (p_k) is log-concave on the integer hull of three ordered, disjoint integer intervals J_L, J_M, J_R of equal cardinality. Then*

$$\sum_{k \in J_M} p_k \geq \min \left(\sum_{k \in J_L} p_k, \sum_{k \in J_R} p_k \right).$$

Proof. In each outer interval select a term at least its interval average. Discrete concavity of $\log p_k$ forces every intermediate term to be at least the smaller of the two selected terms. Sum this inequality over the middle interval. □

Set $d_h = |D_h| \rightarrow \infty$ and $K_h = \lfloor d_h/4 \rfloor - 2$. Take equal-length integer intervals of radius K_h centered at integers nearest to μ_h^0, μ_h^1 , and $(\mu_h^0 + \mu_h^1)/2$. For large h these are ordered and disjoint, with the midpoint interval in between. Call the outer intervals J_L, J_R and the middle interval J_M . Chebyshev's inequality and Lemma 2 imply

$$\begin{aligned} \min\{\mathbb{P}(X_h \in J_L), \mathbb{P}(X_h \in J_R)\} &\geq \min(q, 1-q) \left(1 - \frac{5}{16} - o(1) \right) \geq \frac{539}{1600} - o(1), \\ \mathbb{P}(X_h \in J_M) &\leq \frac{5}{16} + o(1) = \frac{500}{1600} + o(1). \end{aligned}$$

For the first bound, the appropriate conditional mean is at distance at least $d_h/4 - O(1)$ from the complement of its interval. For the second, the middle interval is at least that distance from either conditional mean. The strict gap $39/1600$ proves that the law of X_h is not log-concave on the hull of these intervals.

This transfers exactly to the unweighted independence coefficients, since

$$\mathbb{P}(X_h = k) = \frac{i_k(T_h)\lambda^k}{I(T_h; \lambda)}, \quad p_k^2 - p_{k-1}p_{k+1} = \frac{\lambda^{2k}}{I(T_h; \lambda)^2} (i_k^2 - i_{k-1}i_{k+1}). \quad (7)$$

Thus there is a strict log-concavity failure at some interior index k_h in this hull. If $M_h = \mathbb{E}X_h$, its location satisfies

$$k_h = M_h + O(d_h). \quad (8)$$

The support and the asymptotic location of these intervals are checked next.

4. The failure stays a positive fraction from the endpoint. Let p_ℓ be the occupation probability of a vertex at level $\ell \leq h$ in the full tree. Conditional on an unoccupied parent this probability is q , and on an occupied parent it is zero. Hence

$$p_0 = q, \quad p_{\ell+1} = q(1 - p_\ell), \quad p_\ell = \pi + (q - \pi)(-q)^\ell, \quad \pi = \frac{q}{1+q}.$$

A terminal vertex on level h and its extra leaf contribute expected size $r + (1 - r)p_h$. Therefore

$$\frac{M_h}{b^h} = \sum_{\ell=0}^{h-1} b^{\ell-h} p_\ell + r + (1 - r)p_h \longrightarrow L := \frac{\pi}{b-1} + r + (1 - r)\pi.$$

The convergence of the sum follows by splitting it at a fixed distance from its last term and bounding the remaining geometric tail.

To calculate $\alpha_h = \alpha(T_h)$, let u_h, v_h be the maximum sizes conditional on root state 0, 1. Then

$$u_0 = v_0 = 1, \quad u_{h+1} = b \max(u_h, v_h), \quad v_{h+1} = 1 + bu_h.$$

Their difference alternates between 1 at odd heights and $1 - b$ at positive even heights. Thus

$$\alpha_h = b^h + \sum_{j=0}^{\lfloor (h-1)/2 \rfloor} b^{h-1-2j}, \quad \frac{\alpha_h}{b^h} \longrightarrow A := 1 + \frac{b}{b^2 - 1}.$$

Since $0 < r < 1$ and $\pi < 1/3$,

$$0 < L < 1 + \frac{1}{3(b-1)} < A.$$

Together with $d_h = o(b^h)$, this also shows that all three intervals above lie strictly inside the support for sufficiently large h . Equations (8) and (1) now follow, and

$$\beta = \frac{L}{A} < \frac{1 + 1/189}{1 + 64/4095} = \frac{12350}{12477} < \frac{99}{100}.$$

This completes the proof of Theorem 1. □

A further consequence. The total variance is at least $q(1 - q)D_h^2$. The exact formula for D_h shows $|D_h| \asymp \rho^h$: its growing-mode coefficient is nonzero, since $\delta \leq b^{-2} < 1/(1 + \rho)$. Hence

$$\text{Var } X_h \geq c_0 \rho^{2h}, \quad |V(T_h)| \asymp b^h, \quad \rho^2 > b.$$

At this fixed activity the variance is superlinear in the number of vertices. This does not conflict with the activity interval used for the central limit theorem in [2]; our activity lies between $2^{63} - 1$ and $2^{64} - 1$.

Exact finite verification. The accompanying Python program uses only rational interval arithmetic, enclosing δ in $[2^{-64}, 2^{-63}]$ and propagating (4)–(5). It certifies a finite instance at height 8:

$$|V(T_8)| = 567417810194497, \quad \alpha(T_8) = 285874097225792.$$

All three equal-length windows lie in the integer interval

$$[282933720312059, 282983698113316],$$

whose upper endpoint is strictly below $0.99\alpha(T_8)$. Both outer-window probabilities exceed 0.37, whereas the middle-window probability is below 0.26. These rounded bounds are asserted against exact rational enclosures by the verifier. Lemma 3 therefore certifies a failure inside this explicit interval without constructing the enormous coefficient list. The script separately checks the conditional-moment identities against exact polynomial calculations on small trees. These checks supplement the all-heights argument above.

Research status. This is a newly derived, unrefereed proof draft. No unproved analytic lemma is used in its argument. The proposed answers to the cited open questions depend on the correctness of this draft; they should be independently checked before a priority or publication claim is made. Literature checked through 27 September 2026 includes the sources below. No resolution of the all-sizes unimodality conjecture is claimed.

References

- [1] D. Galvin, *Trees with non log-concave independent set sequences*, arXiv:2502.10654v2, Question 3.1. <https://arxiv.org/abs/2502.10654>.
- [2] E. X. Fang, J. Lu, E. Nevo, Y. Yao and H. Zheng, *Unimodality of Independence Polynomials for Sufficiently Large Forests*, arXiv:2609.20961v1, 17 September 2026, Theorem 1.2 and Section 9. <https://arxiv.org/abs/2609.20961>.
- [3] X. Yu, *Unbounded runs of non-log-concavity in independence polynomials of trees*, manuscript in reproducibility archive, version 1.1.0, 3 September 2026. <https://doi.org/10.5281/zenodo.22263331>.